

GNN-Based BERT for Understanding Context from Music

Course: Neural Networks (CSE425 / EEE474 / CSE715)

Prepared By: Tanisha Islam

Instructor: Moin Mostakim

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Abstract

Music contains both acoustic structure and semantic context. This project investigates whether graph neural networks and transformer-based language representations can be combined for music-context understanding. Audio is resampled to 22,050 Hz and represented using 128-bin log-mel features, chroma, and MFCC-derived segment features. Temporal adjacency and acoustic cosine similarity are used to construct music structure graphs. GraphSAGE and GAT-style graph processing are evaluated together with DistilBERT text representations. The project includes BERT-only, CNN, GNN-only, early concatenation, and cross-attention fusion experiments. A separate dual-encoder contrastive model is evaluated on genuine MusicCaps audio-caption pairs using InfoNCE loss and retrieval Recall@K.

Experiments were conducted using 1,000 MagnaTagATune examples divided into 800 training, 100 validation, and 100 test clips with a 50-tag vocabulary. MusicCaps contributed 188 clean audio-caption pairs for the contrastive experiment. The strongest tag-ranking result was obtained by the CNN baseline with AUC-PR 0.2337, while fixed-threshold Macro-F1 values remained low because of strong multi-label imbalance and limited training. The contrastive model achieved $R@1 = 0.05$, $R@5 = 0.30$, and $R@10 = 0.45$. These results are reported without claiming state-of-the-art performance.

Keywords: GNN, BERT, music information retrieval, multimodal learning, GraphSAGE, cross-attention, contrastive learning

1. Dataset and Preprocessing

1.1 MagnaTagATune

The primary supervised tagging experiments use 1,000 real MagnaTagATune clips. The final subset contains 800 training, 100 validation, and 100 test samples. Fifty commonly occurring music tags form the multi-label prediction vocabulary. The official split information is preserved where available in the preprocessing pipeline.

1.2 MusicCaps

The cross-modal task uses 188 genuine MusicCaps audio-caption pairs. These are real paired examples rather than synthetic combinations of unrelated audio and captions. For contrastive training, the implementation uses 150 training, 18 validation, and 20 test examples.

1.3 Audio Processing

Audio is resampled to 22,050 Hz. The preprocessing module extracts normalized 128-bin log-mel spectrograms together with chroma and MFCC information. Audio is divided into fixed-duration segments. Each segment becomes a node in the graph and is represented by a 32-dimensional acoustic feature vector.

1.4 Graph Construction

Two edge types are created. First, temporal edges connect adjacent audio segments. Second, similarity edges connect acoustically related segments when cosine similarity exceeds the configured threshold. This provides a graph representation capable of connecting repeated or similar regions even when they are not directly adjacent in time.

1.5 Text Processing

Text is encoded using HuggingFace DistilBERT with a maximum token length of 128. For MagnaTagATune, the text context is derived from available title and artist metadata rather than from the target tags themselves, which avoids direct label leakage. MusicCaps uses its genuine expert-written captions.

2. Model Architectures

2.1 CNN Mel-Spectrogram Baseline

A convolutional neural network operates directly on real log-mel spectrograms. It provides an audio-only baseline against which graph-based representations can be compared.

2.2 Task 1: BERT-only

DistilBERT converts text context into contextual token embeddings. A classification head predicts the 50 music tags using binary cross-entropy loss. The pretrained language backbone is frozen by default to reduce computational cost.

2.3 Task 2: GNN-only

GraphSAGE is applied to the segment graph. Message passing combines each node with information from neighboring temporal and similarity-connected segments. Global graph pooling produces a fixed-size audio representation for multi-label classification. The codebase also contains GAT support.

2.4 Task 3: Multimodal Fusion

Two multimodal variants are evaluated. Early concatenation joins pooled graph and text representations directly. The cross-attention model instead lets the graph representation attend to contextual BERT token embeddings before classification. Both variants predict only multi-label music tags in the final implementation.

2.5 Task 4: Contrastive Audio-Text Alignment

A dual-encoder model maps the audio graph and MusicCaps caption into a shared embedding space. InfoNCE loss encourages matched audio-caption pairs to be closer than mismatched examples. The resulting representation is evaluated using text-to-audio retrieval and zero-shot tag prompts.

3. Experimental Results

Model	Macro-F1	AUC-PR
Random Tags	0.0575	0.0957
Majority Tags	0.0000	0.0637
CNN Mel-Spec	0.0239	0.2337
Task 1: BERT-only	0.0000	0.1024
Task 2: GNN-only	0.0091	0.1907
Task 3: Early Concat	0.0000	0.0887
Task 3: Cross-Attention	0.0000	0.0935
Task 4: Zero-Shot	0.0408	0.0634

The CNN baseline produced the highest AUC-PR value of 0.2337. GraphSAGE obtained AUC-PR 0.1907. BERT-only, early concatenation, and cross-attention produced lower ranking performance in this limited experiment. The low Macro-F1 scores, including zero for several models at the fixed decision threshold, show that the current training setup does not provide strong calibrated multi-label classification.

Importantly, the random baseline achieved Macro-F1 0.0575, which is greater than several trained models at the fixed threshold. Therefore, the results should not be interpreted as evidence that the fusion architecture outperforms all baselines. AUC-PR provides a more informative view of ranking ability under the severe class imbalance present in the 50-tag setting.

3.1 MusicCaps Retrieval

Metric	Result
Recall@1	0.0500
Recall@5	0.3000
Recall@10	0.4500

4. Visual Analysis

Figure 1. Training losses and available validation performance trends.

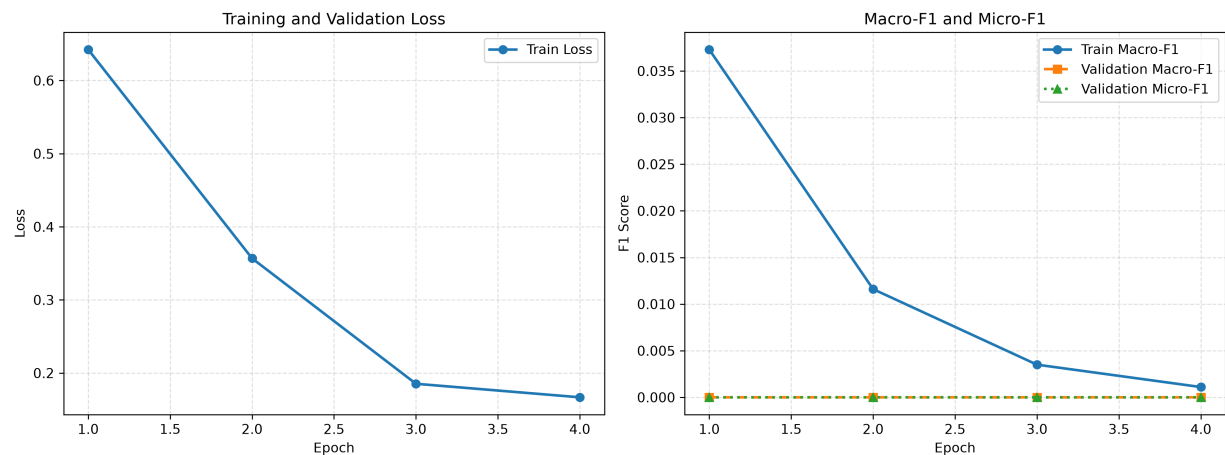


Figure 2. t-SNE visualization of learned multimodal representations.

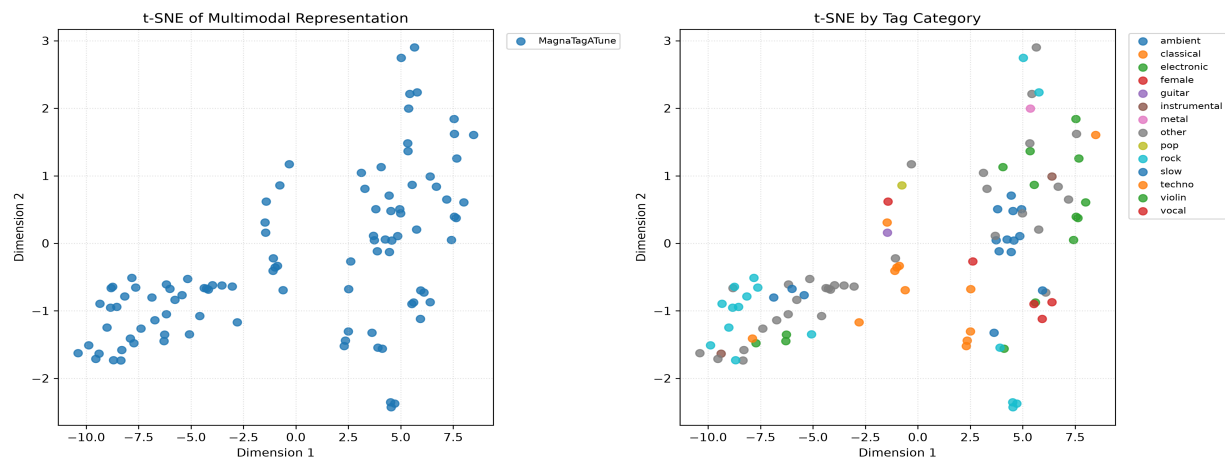
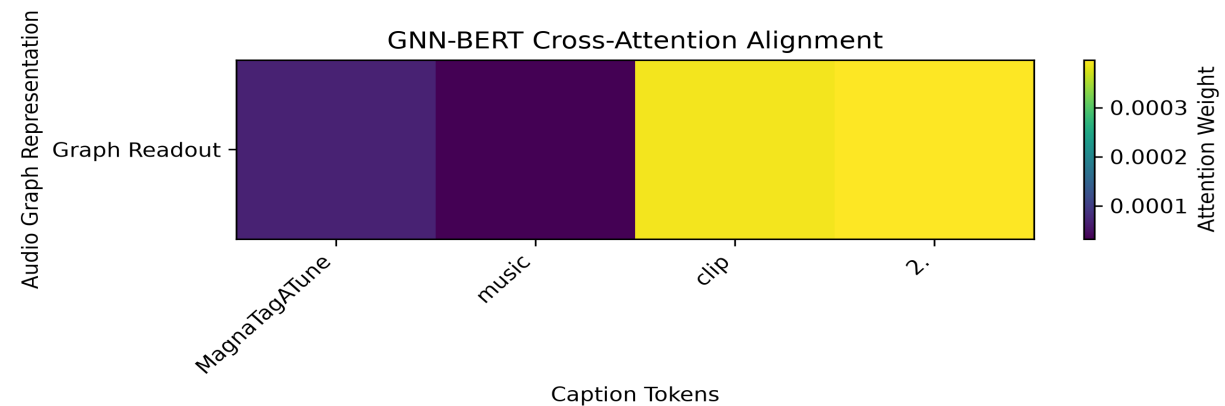


Figure 3. Cross-attention visualization between the audio graph representation and text tokens.



5. Qualitative Retrieval Case Studies

Case Study 1: Synth / Electronic Query

A query describing synthesizer-heavy electronic music retrieved examples with mixed relevance. Some top-ranked clips were mismatches, while another result showed broader electronic and pop production overlap. This case demonstrates that the learned embedding contains useful coarse semantic structure but does not consistently produce exact semantic matches.

Case Study 2: Pop Production Query

For a pop-oriented production description, the matching caption was retrieved at rank 2. This provides a positive example in which the contrastive representation placed the corresponding audio close to its textual description, although it did not achieve the top rank.

Case Study 3: Hindustani Classical Query

For a Hindustani classical description, a sitar-based meditation example was retrieved at rank 1. The match shares broad cultural, instrumental, and acoustic characteristics with the query. This example suggests that the model can capture some high-level instrument and style associations even with a small training set.

Interpretability

The cross-attention visualization provides a qualitative indication of which text tokens interact most strongly with the pooled graph representation. These maps should be interpreted as model attention patterns rather than causal explanations. The project does not contain lyric annotations, so no claim is made about lyric-level alignment.

6. Limitations and Deviations

Task 2 dataset deviation: The project specification proposes genre classification on GTZAN or FMA-small. Because of the limited project duration and computational constraints, the implemented Task 2 uses the available MagnaTagATune multi-label tagging setup instead. Therefore, the GNN-only experiment should be interpreted as audio-graph tag classification rather than the requested GTZAN/FMA genre experiment.

Task 3 qualitative limitation: The dataset used in the final pipeline contains no lyrics. Consequently, the required graph-path-plus-lyric alignment analysis could not be completed literally. The project instead provides graph/text cross-attention visualization and caption-based qualitative analysis.

Training scale: Models were trained for only four epochs on a reduced subset of MagnaTagATune. The MusicCaps contrastive experiment contains only 188 clean pairs. These choices made the project computationally manageable but strongly limit generalization and model convergence.

Threshold sensitivity and imbalance: The multi-label target distribution is highly imbalanced. A single fixed probability threshold produces very low Macro-F1 for several models even when their AUC-PR indicates some ranking ability. Future work should tune per-class thresholds, use class-balanced losses, train for more epochs, and evaluate larger datasets.

7. Conclusion

This project implements an end-to-end multimodal music-context pipeline combining real audio preprocessing, graph construction, GraphSAGE/GAT components, DistilBERT text representations, multimodal fusion, and contrastive audio-text alignment. Although the experimental results are modest, the implementation demonstrates the full technical workflow required to investigate graph-based and language-based representations of music.

The strongest supervised ranking result was produced by the CNN mel-spectrogram baseline with AUC-PR 0.2337. The MusicCaps contrastive model achieved Recall@1 of 0.05, Recall@5 of 0.30, and Recall@10 of 0.45. These results establish a reproducible baseline for future work rather than a state-of-the-art claim.

8. Project Deliverables

The repository includes the complete source code, configuration, dataset preparation scripts, training and evaluation modules, representative graph JSON files, evaluation metrics, plots, qualitative retrieval examples, an exploratory notebook, and notebooks/demo_context.ipynb containing an end-to-end multimodal forward-pass demonstration.

Repository: github.com/tanishaislam-eng/gnn-bert-music-context